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Dilute Electrolyte with Vehicular Aggregates for Stable High-Energy Lithium-Metal Batteries

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Electrolyte design is critical for high-energy lithium-metal batteries (LMBs) because it dictates the aggressive Li–electrolyte interphase that governs cycling stability and Coulombic efficiency (CE). However, traditional high- or locally high-concentration electrolytes, which achieve good Li-metal compatibility by the formation of anion–solvent–Li^+ aggregates (AGG), typically suffer from poor ionic conductivity (e.g., ≈1 mS cm^{-1}). Therefore, achieving both high ionic conductivity and AGG-dominated solvation structure under dilute conditions still remains a challenge. Herein, a novel dilute electrolyte with vehicular aggregates (DVA) mechanism is proposed by employing pyrrolidine-1-sulfonyl fluoride (PSF) solvent with optimal steric hindrance effect. Based on this DVA electrolyte, the uniquely AGG-dominated solvation structure under dilute conditions achieve high ionic conductivity (4.9 mS cm^{-1}) featuring its vehicular ion-transport mechanism. It exhibits good Li-metal compatibility with high stripping-plating CE of ≈99.5% with inorganic (LiF, Li_2O)-rich robust solid–electrolyte interphase. Furthermore, the electrolyte effectively suppresses the stress-corrosion cracking, transition-metal dissolution, gas evolution, and detrimental surface degradation on the cathode side, thus enabling 4.6-V-class LMBs with ultra-high-Ni cathodes to deliver a high discharge capacity of 228.4 mAh g^{-1}, excellent rate capability up to 2C, and 87% capacity retention after 150 cycles. This work offers a promising approach for designing advanced electrolytes for high-energy LMBs.

1. Introduction

Lithium-metal anodes (LMAs) have been considered as a promising candidate for high-energy batteries due to their high theoretical specific capacity (3862 mAh g^{-1}) and low potential (−3.04 V vs. Standard Hydrogen Electrode).[1] However, their high reactivity has greatly hindered the practical applications due to the parasitic side reactions between LMAs and electrolyte and accompanying uncontrollable Li-dendrite growth, thus significantly deteriorating the reversibility and cyclability of Li-metal batteries (LMBs).[2,3] Designing a suitable electrolyte to stabilize the Li–electrolyte interphase has recently received great attention to achieve highly reversible and stable LMBs.

The basic logic for electrolyte design is to tailor the competitive reactivity of solvent molecules and salt anions at the interface with LMAs. Such competitive reactivity is related to both the driving force for reduction and the collision probability with the Li-metal surface. In a conventional dilute electrolyte with ethylene carbonate (EC) as the primary polar solvent, Li^+ cations are fully solvated by EC molecules, forming solvent solvent-separated ion pairs (SSIPs). In this scenario, ion transport is dominated by the "vehicular" mechanism,[4] wherein Li^+ cations migrate together with their solvation sheaths (**Figure 1a**, inset), resulting in a high ionic conductivity σ of ≈10 mS cm^{-1} and a low transference number (t_{Li+}) of <0.5. Near the Li-metal surface, within the inner Helmholtz plane (IHP),[5] the relative probability density of solvent molecules is high due to their co-transport with Li^+ cations, which are electrostatically attracted to the negatively charged Li-metal surface (Figure 1a). Consequently, the solid-electrolyte interphase (SEI)[6] formed in this electrolyte is primarily derived from solvent decomposition, which is unfavorable in LMBs as most non-fluorinated carbonates have severe side reactions with Li-metal.[7] To facilitate the anion-derived SEI[8] and thus improve the Li-metal compatibility, high concentration electrolytes (HCEs)[9–12] or localized high concentration electrolytes (LHCEs)[13–16] have been developed to promote the formation of coordinated ion pairs (CIP) and anion–solvent–Li^+ aggregates (AGGs). These clusters allow anions to migrate along with

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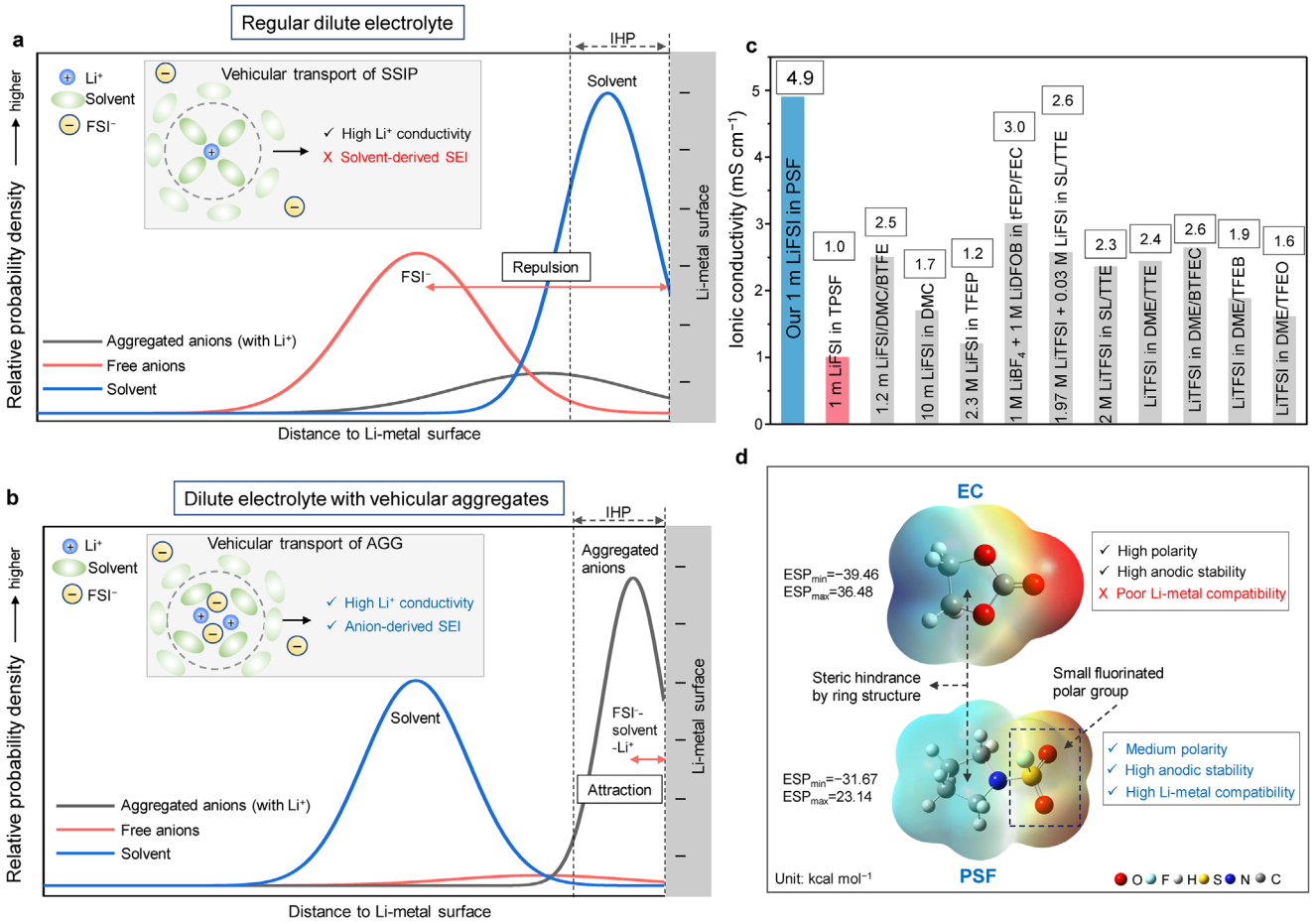


Figure 1. Design principle for the DVA electrolyte. Schematic illustration of the probability distribution of electrolyte components near Li-metal surface in a regular dilute electrolyte a) and dilute electrolytes with vehicular aggregates (DVA electrolyte, b). In a regular dilute electrolyte, anions (e.g., FSI^-) are usually repelled from the negatively charged Li-metal surface, reducing their probability of collision and subsequent reaction with the Li-metal surface. Consequently, the formation of SEI is solvent-dominated. In the DVA electrolyte, aggregated anions (coordinated with Li^+ and solvent molecules) are attracted toward the Li-metal surface, thereby increasing their probability of collision and interfacial reaction, ultimately leading to the formation of anion-derived SEI. c) Ionic conductivities of the DVA electrolyte and other reported electrolytes dominated by AGGs: 1 m LiFSI in 1-((trifluoromethyl)sulfonyl)pyrrolidine (1.0 mS cm^{-1}, TPSF),[22,23] 1.2 m LiFSI/dimethyl carbonate (DMC)-bis(2,2,2-trifluoroethyl)ether (BTFE)[13] (2.5 mS cm^{-1}), 10 m LiFSI/DMC (1.7 mS cm^{-1}),[12] 2.3 M LiFSI/tris(2,2,2-trifluoroethyl)phosphate (TFEP)[24] (1.2 mS cm^{-1}), 1 M $LiBF_4$+1 M lithium difluoro(oxalato)borate (LiDFOB) in ethyl 3,3,3-trifluoropropanoate (tFEP)-fluoroethylene carbonate (FEC)[5] (3.0 mS cm^{-1}), 1.97 M LiTFSI + 0.03 M LiFSI in sulfolane (SL)/1,1,2,2-tetrafluoroethyl-2,2,3,3-tetrafluoropropyl ether (TTE) (2.6 mS cm^{-1}), 2 M LiTFSI in SL/TTE (2.3 mS cm^{-1}),[25] 1.06–1.65 M LiTFSI in 1,2-dimethoxyethane (DME) with different diluent solvents TTE (2.4 mS cm^{-1}), bis(2,2,2-trifluoroethyl) carbonate (BTFEC) (2.6 mS cm^{-1}), tri(2,2,2-trifluoroethyl) borate (TFEB) (1.9 mS cm^{-1}) or tris(2,2,2-trifluoroethyl) orthoformate (TFEO) (1.6 mS cm^{-1}).[26] Our DVA electrolyte has the higher ionic conductivity (4.9 mS cm^{-1}) among them. d) Electrostatic potential (ESP) distribution maps of ethylene carbonate (EC) and pyrrolidine-1-sulfonyl fluoride (PSF) evaluated by DFT calculations. A similar alkyl ring structure is adopted in PSF from EC to introduce steric hindrance, which helps modulate the coordination with FSI^- anions. A small fluorinated polar group ($-FSO_2$) is incorporated in PSF to generate a more delocalized negative electrostatic potential, facilitating lithium salt dissociation with moderate polarity.

Li^+
toward the Li-metal surface,[17] where electrostatic attraction leads to their interfacial enrichment (the probability density of anions also shifts toward Li-metal surface, Figure 1b) and subsequent decomposition, forming a protective SEI. It often results in a higher t_{Li+}>0.5 but a low σ<3 mS cm^{-1} (Figure 1c), where ion transport is governed by a "hopping" or hybrid "hopping + vehicular" mechanism. Therefore, balancing the trade-off between high ionic conductivity and anion-derived SEI formation remains a key challenge toward realizing highly reversible LMBs.

Recently, a lot of great works have inspired a broad range of novel solvation structures. Ren et al. proposed a molecular anchoring approach to tune the coordination of free ether solvent molecules in LHCEs, which suppressed the excessive side reactions.[14] Xu et al. successfully introduced a new long-chain ethylene glycol di-*n*-butyl ether to further condense the Li^+ solvation sheath and form compact ion-pair aggregates in LHCEs[15] for high-energy LMBs. Suo et al. proposed a new tri-fluorinated linear ester (ethyl 3,3,3-trifluoropropanoate) to collaborate with fluoroethylene carbonate to enrich the number of anions within the IHP.[5] Fan et al. pioneered a ligand-channel transport

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mechanism enabled by significantly reducing the size of the Li^+ solvation shell by a small fluoroacetonitrile solvent in ultrafast charging graphite-based Li-ion cells.[4] Kang et al. reported a rationally designed ionic plastic crystal into conventional electrolyte to form a miniature anion-Li^+ solvation structure for practical and safe LMBs.[18] Deng et al. designed a trifluoromethylation-modified ether molecule with solvation structure via lithiophobicity, leading to superior interphasial chemistries in layered structure and fast Li^+ transport kinetics.[19] Despite these advances, the development of electrolytes that simultaneously enable high ionic conductivity, transference number, and Li-metal compatibility remains elusive, highlighting the need for rational solvent design featuring both strong interfacial stability and AGG-forming capability.

In the present work, we report the design of a 1 mol kg^{-1} single-solvent, single-salt dilute electrolyte featuring an AGG-dominated solvation structure and vehicular ion transport, a synergistic configuration that, to the best of our knowledge, has not been explicitly articulated in prior work despite occasional observations of abundant AGG species in related systems.[20] For example, the formation of AGGs relies on high salt concentration with a single solvent, where a low solvent-to-salt molar ratio is achieved to force solvent molecules to be shared by anions. Otherwise, a binary solvent mixture composed of polar and nonpolar components is required to promote anions' evasion into the Li^+ solvation shell. However, it is challenging for dilute (≈1 m) and single-solvent conditions to form a lot of AGGs. To achieve this, the key is to suitably tailor the solvent's polarity by molecular design. On one hand, we designed a highly Li-metal compatible polar end (FSO_2–) with a moderately delocalized negative charge centered on the FSO_2– moiety (Figure 1d).[20,21] This balanced charge distribution is crucial: an overly localized negative charge (–C═O in EC molecule) would result in strong Li^+–solvent coordination and thus promote the formation of SSIPs. Conversely, excessive delocalization (e.g., $CF_3SO_2^-$,[22,23] 1-((trifluoromethyl)-sulfonyl pyrrolidine, TPSF) leads to poor salt dissociation that compromises the ionic conductivity (≈1 mS cm^{-1}, as compared to our electrolyte, 4.9 mS cm^{-1}, Figure 1c). On the other hand, inspired by EC's molecular structure, a five-membered ring structure (N-pyrrolidine) is introduced to sterically shield the positive charge center to modulate its interaction with the salt anion (Figure 1d). The N-pyrrolidine moiety is typically resistant to reduction, and its oxidative stability is further enhanced by the adjacent electron-withdrawing FSO_2– group. Therefore, a dilute electrolyte 1 m lithium bis(fluorosulfonyl) imide (LiFSI) in pyrrolidine-1-sulfonyl fluoride (PSF) was developed. It is interesting to note that the dominant solvation clusters in this electrolyte are AGGs, which is confirmed by both molecular dynamics (MD) simulations and Raman spectroscopy. The dilute nature successfully leads to a high ionic conductivity of 4.9 mS cm^{-1} (higher than most LHCEs and HCEs) and a transference number t_{Li+}>0.5, suggesting a new dilute electrolyte with vehicular aggregates mechanism (DVA electrolyte for short, Figure 1b inset). Our DVA electrolyte exhibited good compatibility with LMAs to achieve a high Li-metal stripping-plating Coulombic efficiency (CE) of ≈99.5% with inorganic (LiF, Li_2O)-rich robust SEI. Benefiting from its high anodic stability, the 4.6-V-class Li||nickel-rich cathodes with the DVA electrolyte delivered a high discharge capacity of 228.4 mAh g^{-1}, excellent rate capability to 2C, and a high capacity retention of 87% after 150 cycles, which was also verified by a pouch cell under practical conditions. Our DVA electrolyte successfully suppresses the stress-corrosion cracking, transition-metal dissolution, gas evolution, and detrimental surface degradation on the cathode side. This work provides a promising approach for designing new electrolytes in high-voltage LMBs.

2. Results and Discussion

To evaluate the cell performance and interphase evolution in high-voltage LMBs, a conventional dilute electrolyte with the vehicular ion transport mechanism based on EC was selected (1 mol L^{-1} $LiPF_6$ in EC/EMC) as the reference electrolyte (so-called the carbonate-based electrolyte). Basic physicochemical properties, including melting point, viscosity, polarity, and flammability, were first evaluated. Our PSF solvent and DVA electrolyte possesses favorable physicochemical properties—weak polarity (much lower than DME and comparable to EMC, Figures S1 and S2; Table S1 and Note S1, Supporting Information), low viscosity, low melting point, and intrinsic flame-retardancy (Figure S3, Supporting Information)—making it a suitable electrolyte that enhances cell safety. Molecular dynamics (MD) simulations employing nonpolarizable and polarizable force fields (Figures S4–S6, Table S2–S4, Supporting Information) and density functional theory (DFT) calculations were first conducted to simulate the interactions between ions and solvents. DFT calculation of the electrostatic potential (ESP) and surface charge distribution revealed distinct characteristics between the EC and PSF molecules. The EC molecule displays a maximum ESP (ESP_{max}) of 36.48 kcal mol^{-1} and a minimum ESP (ESP_{min}) of −39.46 kcal mol^{-1} (Figure 1d), with the ESP_{min} located on the oxygen atom adjacent to the carbonyl group (C═O). In contrast, the PSF molecule demonstrates a lower ESP_{max} of 23.14 kcal mol^{-1} and a higher ESP_{min} of −31.67 kcal mol^{-1} (Figure 1d), where the negative charge is delocalized by the FSO_2– group. This results in reduced solvating capability, indicating weaker coordination abilities toward Li^+ and promoting the formation of AGG. Binding energies (E_{bind}) of Li^+ with solvents were also calculated (Figure S7, Supporting Information). The E_{bind} of PSF (43.95 kcal mol^{-1}) is comparable to values reported for the weakly solvating solvent 1,3-dioxolane (DOL, 38.25 kcal mol^{-1}) and lower than those of common solvents like 1,2-dimethoxyethane (DME, 64.29 kcal mol^{-1}). Furthermore, when Li^+ coordinates with three DME molecules, the E_{bind} (121.18 kcal mol^{-1}) is much stronger than when coordinated with three PSF molecules (86.11 kcal mol^{-1}). These findings are consistent with the characterization of the solvation force of different solvents (Figure S1–S2 and Table S1, Supporting Information), suggesting that PSF belongs to the category of weakly solvating electrolytes. The salt–solvent clusters (SSIP, CIP, and AGG) are classified based on the Li^+–FSI^- coordination number (CN, **Figure 2a**). MD simulations and radial distribution function (RDF, Figure S8, Supporting Information) analyses were conducted for the DVA electrolyte system using PSF as the solvent. LiFSI and PSF form a 3D network of percolating salt–solvent clusters (Figure 2b). Notably, within a simulation time of 50 ns, the DVA electrolyte is predominantly characterized by the solvation structure of AGG (Figure 2c,d; Figure S9, Supporting Information). Raman deconvolution analysis was also used to experimentally quantify the distribution of salt–solvent clusters. It

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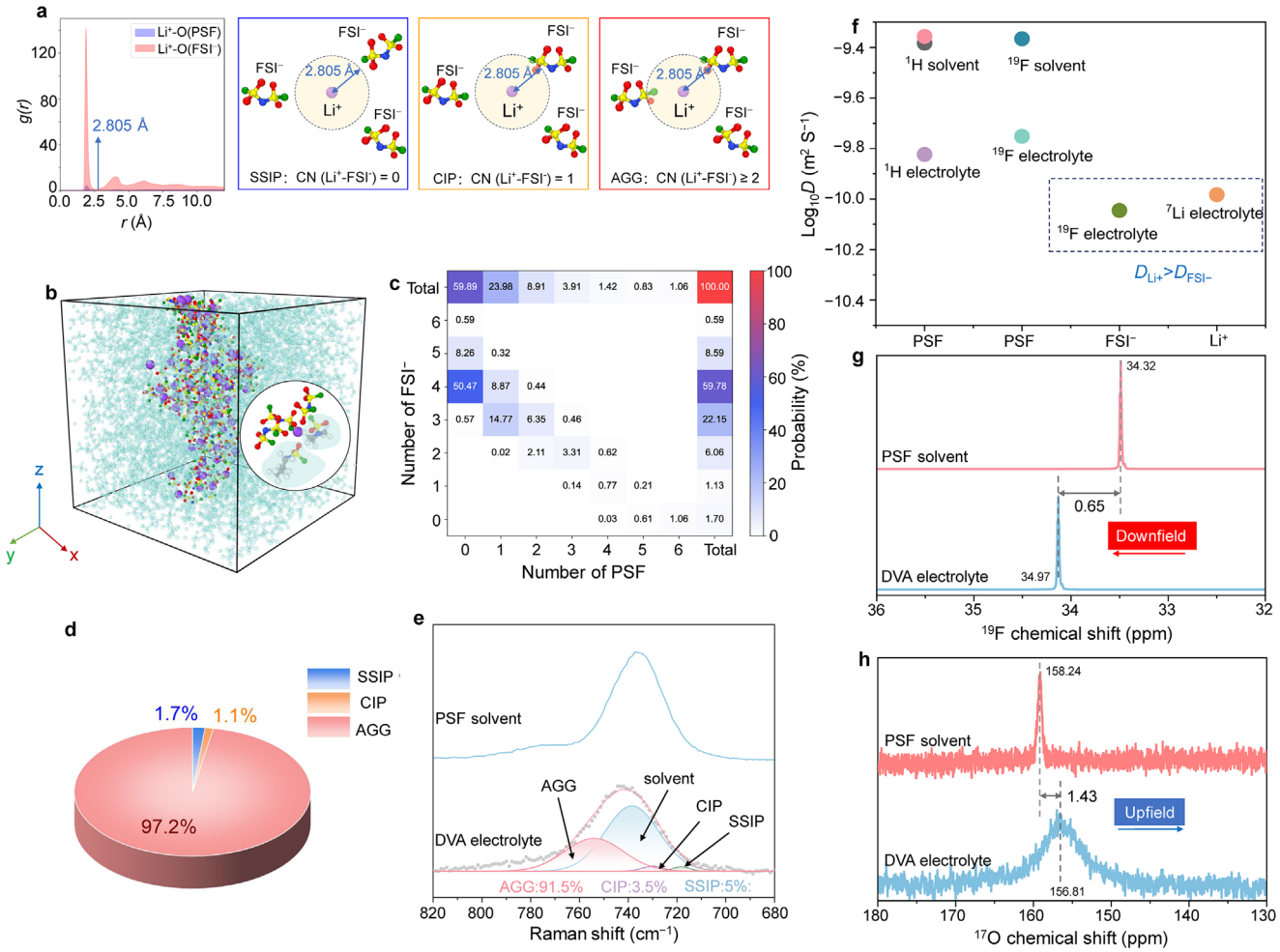


Figure 2. Solvation structure of the DVA electrolyte. a) Definitions of different cluster types (SSIP, CIP, and AGG) by coordination numbers (CN). b) The equilibrium configuration of the MD simulation system and a representative solvation sheath consisting of one Li^+, three FSI^-, and two PSF molecules. c) The probability distribution of various anion–solvent aggregates surrounding a single Li^+ ion. This distribution provides insights into the local Li^+ solvation environment. d) Percentages of SSIP, CIP, and AGG calculated by MD simulations with the polarizable force field. e) Raman spectra of the pure PSF solvent and DVA electrolyte at 25 °C. Deconvolution of Li^+–FSI^- Raman peaks for the DVA electrolyte with peak fits for different cluster types denoted by arrows. f) Diffusion coefficients (D) evaluated by 2D DOSY NMR spectra of the PSF solvent (^1H and ^{19}F), DVA electrolyte (^1H and ^{19}F), and Li^+ cations/FSI^- anions (^{19}F). The ^{19}F g) and ^{17}O h) NMR spectra of the PSF solvent before and after incorporating 1 m LiFSI. 0.65 ppm (downfield) and 1.43 ppm (upfield) peak shifts were noted from the ^{19}F and ^{17}O NMR spectra, respectively.

is apparent that AGG[27–29] is the dominant cluster type in the DVA electrolyte (Figure 2e), which is consistent with the MD simulations. Such a highly bonded cation–anion characteristic greatly benefits the transference number t_{Li+} by reducing the mobility of anions. A diffusion ordered spectroscopy (DOSY) technique based on nuclear magnetic resonance (NMR) spectroscopy was performed to evaluate the diffusion coefficient (D) of the PSF solvent before and after incorporating LiFSI salt. Upon dissolving 1 m LiFSI, the diffusion coefficient of PSF obviously decreased, as evidenced by the distinct symbols representing the pure solvent and the electrolyte (Figure 2f). Notably, in the DVA electrolyte, the diffusion coefficient of Li^+ (D_{Li+}) is higher than that of FSI^- (D_{FSI-}), indicating a t_{Li+} greater than 0.5, which is also demonstrated by MD simulations (Table S5, Supporting Information). To further reveal the coordination interactions between Li^+ and PSF solvent, the ^{19}F and ^{17}O NMR techniques were utilized. An apparent downfield shift of 0.65 ppm for the fluorine atom in PSF was observed in the DVA electrolyte on dissolving 1 m LiFSI (Figure 2g). For comparison, a more pronounced upfield shift of 1.43 ppm was noted from the ^{17}O nuclei (Figure 2h). These NMR results suggest that the ^{17}O nuclei are directly involved in coordination with Li^+. Under the influence of the strong Coulombic field exerted by Li^+, the electron cloud of the PSF molecule moves toward the center ion Li^+, leading to an increased electron density around the ^{17}O nuclei while partially depleting the ^{19}F nuclei.

The unique solvation structures dominated by AGGs in the DVA electrolyte have great influence on the SEI formation and its compatibility with the LMAs.[30] The Li-metal stripping-plating CEs on a bare copper matrix were evaluated by Li||Cu cells with different electrolytes at 0.5 mA cm^{-2} and 0.5 mAh cm^{-2}. It is noted that the nucleation barrier for Li-metal in body-centered cubic (BCC) form (Li_{BCC}) with the DVA electrolyte is the lowest

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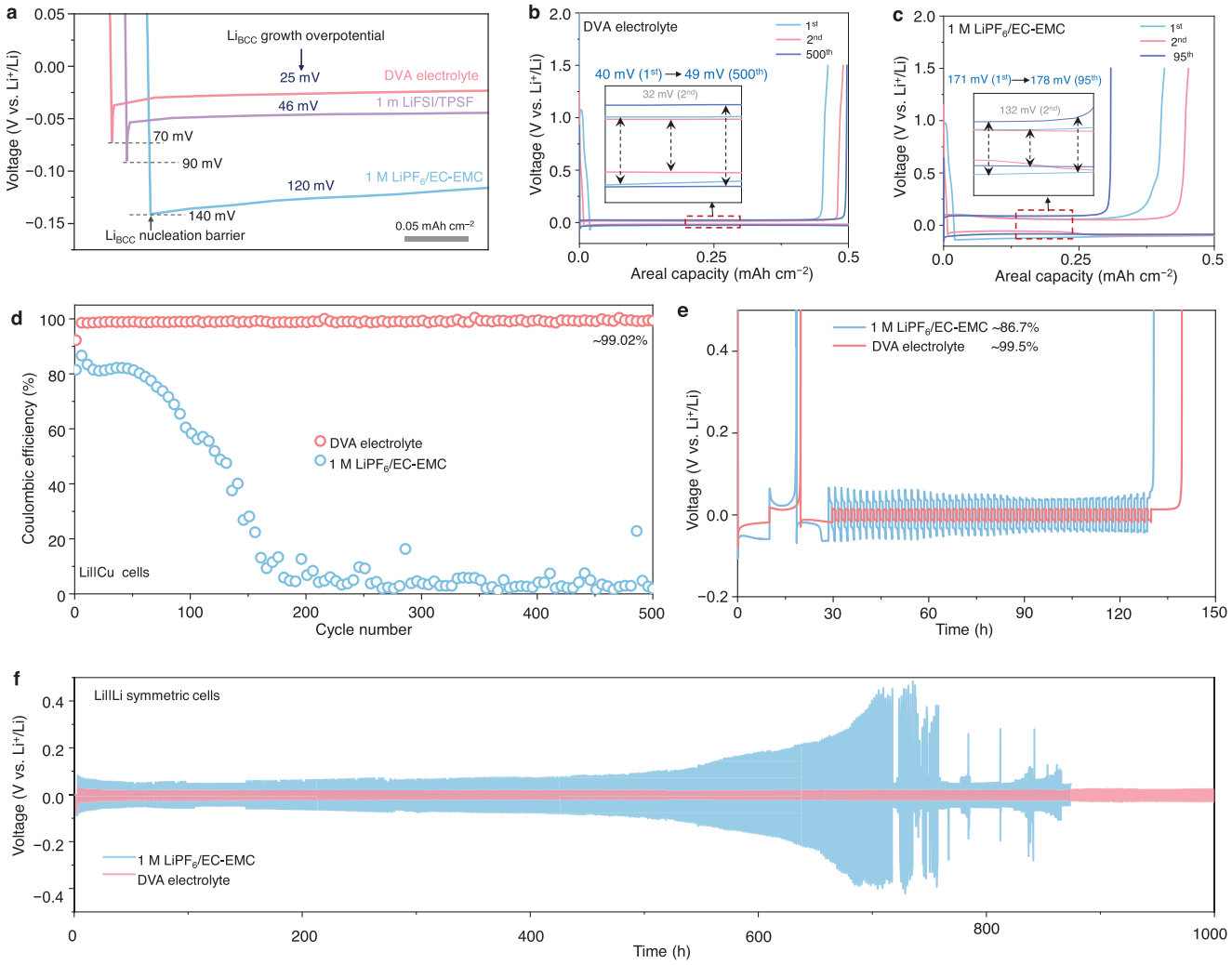


Figure 3. Electrochemical performance evaluated by Li||Cu and Li||Li cells with different electrolytes at 0.5 mA cm^{-2}, 0.5 mAh cm^{-2}. a) The overpotentials during the initial stage of Li deposition in different electrolytes. The DVA electrolyte has the lowest Li_{BCC} nucleation barrier (≈70 mV) and growth overpotential (≈25 mV). Voltage profiles of the Li||Cu cells in the DVA b) and 1 M $LiPF_6$/EC-EMC electrolyte c). d) Li plating/stripping CEs evaluated by Li||Cu cells. e) Average CEs measured by the Aurbach method in different electrolytes. f) Voltage profiles of the Li||Li symmetric cells in different electrolytes.

(70 mV, **Figure 3a**). In contrast, the nucleation barriers with the carbonate reference electrolyte and the TPSF-based electrolyte (1 m LiFSI in TPSF) are higher, at 140 and 90 mV, respectively (Figure 3a). The same trend can be found in the Li_{BCC} growth overpotentials (Figure 3a). The lower overpotentials observed in our DVA electrolyte with PSF as the solvent, compared to the TPSF-based electrolyte, indicate that the smaller fluorinated group (–FSO_2) offers more favorable reaction kinetics than the bulkier –CF_3SO_2 group in TPSF. Such low overpotentials in the DVA electrolyte remained stable upon cycling (Figure 3b) while those of the carbonate-based electrolyte increased obviously (Figure 3c), indicating that the DVA electrolyte can form a stable SEI on the Li-metal surface. The excellent interphasial stability with the DVA electrolyte successfully results in highly reversible LMAs, as evidenced by a high average CE of ≈99.02% over 500 cycles, while the CE decreased rapidly in the carbonate-based electrolyte (Figure 3d). The average CE evaluated by the Aurbach method[31] to exclude the influence of surface impurities on the copper matrix is as high as 99.5% for the DVA electrolyte, much higher than the reference electrolyte (86.7%, Figure 3e). Besides the Li||Cu configuration, Li||Li symmetric cells were further used to evaluate the interphasial stability over cycling (Figure 3f). Voltage profiles of the cell using the DVA electrolyte remain stable over 1,000 h, while those of the carbonate-based electrolyte showed a dramatic increase, further demonstrating that the DVA electrolyte is favorable for highly reversible LMAs. Notably, a negative polarization potential of the first cycle in the DVA electrolyte arises from the native Li_2O/LiOH passivation layer on the Li-metal surface.

In order to elucidate the mechanism for the interphasial stability in the DVA electrolyte, the morphologies of Li plating and SEI compositions in different electrolytes were characterized by scanning electron microscope (SEM), X-ray photoelectron spectroscopy (XPS), and Time-of-flight secondary ion mass

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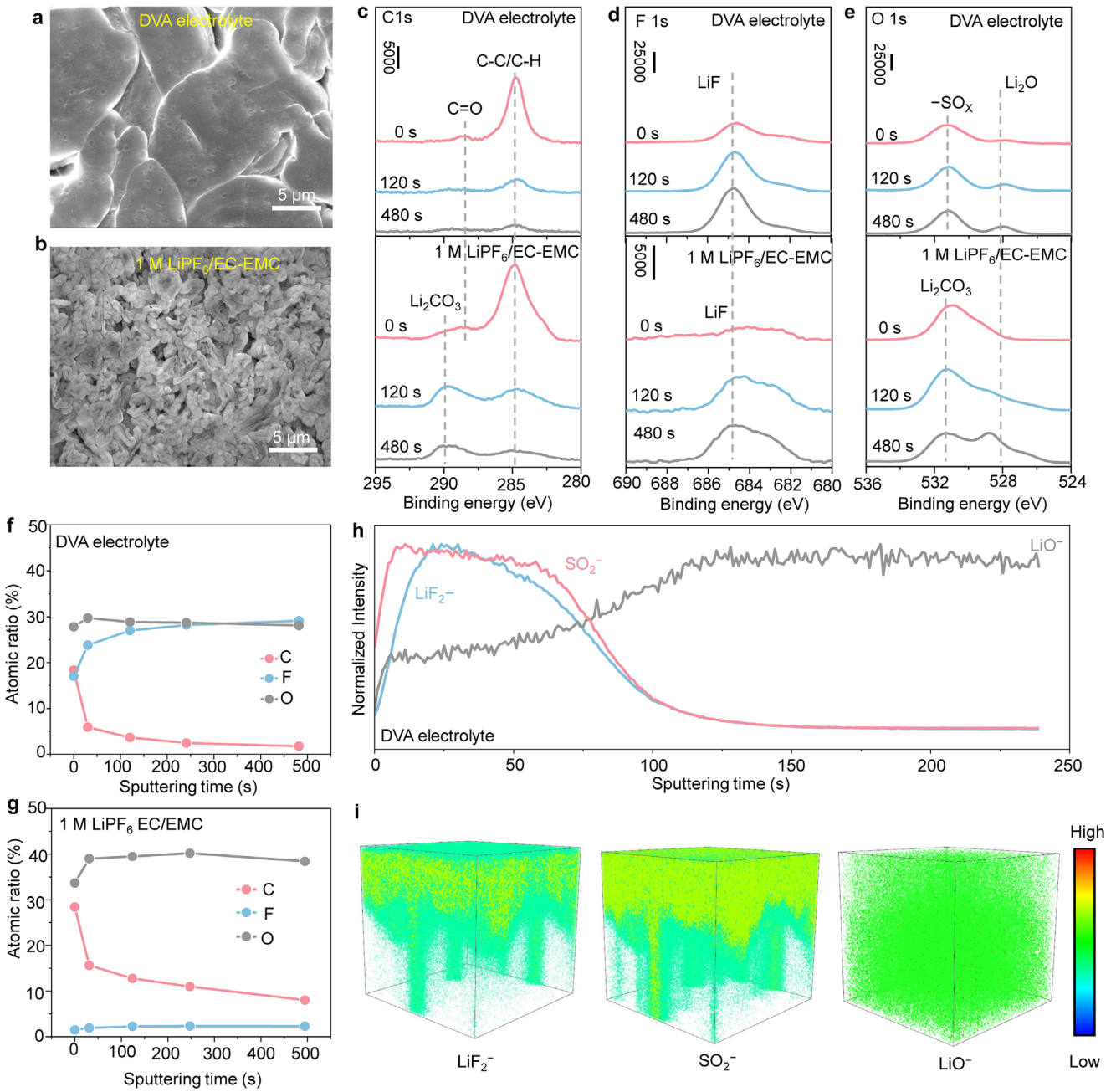


Figure 4. Morphologies and surface chemistry of the SEIs formed on Li metal surface in the DVA and 1 M $LiPF_6$/EC-EMC electrolytes. SEM figures of Li deposits in the DVA a) and 1 M $LiPF_6$/EC-EMC b) electrolytes. C 1s c), F 1s d), O 1s e) XPS spectra of the SEIs formed in both electrolytes and elemental compositions f,g) during Ar^+ sputtering. TOF-SIMS 2D h) and 3D i) distributions of LiF_2^-, $C_2H_3O^-$, and LiO^- from SEIs formed in different electrolytes.

spectrometry (TOF-SIMS). 3 mAh cm^{-2} Li-metal was first deposited on a copper foil at a current density of 0.5 mA cm^{-2}. In the DVA electrolyte (**Figure** 4a), large and chunky Li deposits were observed, whereas needle-like and porous morphologies were formed in the carbonate-based electrolyte (Figure 4b). The smaller surface area of the Li deposits helps suppress side reactions with the electrolyte by reducing the reactive interface, which contributes to the excellent stability observed with the DVA electrolyte. Moreover, to investigate the evolution of SEI composition with different depths, XPS with Ar^+ sputtering was conducted on the Li-metal deposited from both electrolytes. Upon sputtering, the carbonous species in the DVA-derived SEI gradually disappeared, indicating that the inner SEI beneath the Li-metal surface should be composed of inorganic components (Figure 4c). In contrast, for the SEI formed in the carbonate-based electrolyte, the intensity of Li_2CO_3-contained species increased with sputtering depth, suggesting preferential decomposition of EC or EMC solvents and the formation of an organic-rich inner SEI (Figure 4c). The inorganic-rich SEI derived from the DVA electrolyte is dominated by LiF (Figure 4d) and Li_2O

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(Figure 4e),[32] and their combined content within the SEI (Figure 4f) is much higher than that in the SEI formed in the carbonate-based electrolyte (Figure 4g). The low LiF content could be attributed to the SSIP-dominated Li^+ solvation structure in the carbonate-based electrolyte, in which PF_6^- anions are propelled from the negatively charged Li-metal surface, thus reducing their participation in interfacial decomposition and LiF formation. For the SEI formed in the DVA electrolyte, $-SO_x$ and Li_2S (Figure 4e; Figure S10a, Supporting Information) species are also observed, which originate from the cleavage of the S—F bond in FSI^- and/or PSF solvent during the electrochemical reduction process, accompanied by the generation of Li_2O.[33] Specifically, the electrostatic attraction between the Li-metal surface and AGGs leads to localized enrichment of anion probability density at the Li metal/electrolyte interface. To further understand the spatial distribution of SEI chemistry, TOF-SIMS technique was conducted on the Li deposits. The sputtering depth profiles (Figure 4h) and 3D reconstruction images (Figure 4i; Figure S10b, Supporting Information) indicate that the SEI in DVA electrolyte is dominated by LiF_2^--, LiO^--, and LiS-rich inorganic fragments, derived from the decomposition of the FSI^- and/or PSF solvent, which are consistent with the XPS results. The more inorganic components in the DVA-derived SEI lead to better mechanical properties to withstand the high stress concentration during Li-metal plating and stripping (Figure S11, Supporting Information).

Electrochemical performance of high-voltage LMBs with both electrolytes was evaluated by a Li||$LiNi_{0.88}Co_xMn_yO_2$ ($x+y=0.12$, denoted as NCM88) configuration with an upper cut-off voltage of 4.6 V vs. Li/Li^+ at a discharging rate of 0.5 C. Upon charging to a high cut-off voltage of 4.6 V, the Li||NCM88 cells delivered high initial discharging capacities of 228.4 mAh g^{-1} (**Figure 5a**) and 222.8 mAh g^{-1} (Figure 5b) at 0.1C for the DVA and carbonate-based electrolytes, respectively. However, when the rate exceeded 0.1C, a distinct difference in capacity was observed. The cell with the DVA electrolyte delivered specific capacities of 219.8, 212.6, 200.5, 190.2, and 169.2 mAh g^{-1} at 0.1, 0.2, 0.5, 1.0, and 2.0 C, respectively (Figure 5c). In sharp contrast, the cell with the carbonate-based electrolyte only exhibited 183.4, 186.6, 143.4, 89.5, and 13.0 mAh g^{-1} at the same rates (Figure 5c). For the cell with the carbonate-based electrolyte, the cell capacity decreased rapidly in the first 25 cycles and then abruptly failed in several cycles. For this cell, the rapid capacity decay is likely attributed to the severe side reactions at such a high voltage, which can also be reflected by the low CEs ≈90% during the initial five cycles (Figure 5d middle). In strong contrast, a significantly improved cycling performance was noted for the cell using our DVA electrolyte, retaining more than 87% of the original capacity with both high CEs (average ≈99.8%) and energy efficiencies (average ≈97.2%) during the entire cycles (Figure 5d). It indicates that our DVA electrolyte, especially the PSF solvent, has a great anodic stability against the aggressive oxidation chemistry of the Ni-rich cathode at an extreme voltage. The high energy efficiencies also indicate a fast ion transport both in the electrolyte and within the cathode–electrolyte interphase (CEI).[34] In contrast, the 1 m LiFSI/TPSF electrolyte, featuring TPSF's $—CF_3$ group, exhibits lower energy efficiency and poorer cycling stability and rate performance (Figure S12, Supporting Information), which can be attributed to the high impedance on the Li-metal side (Figure 3a). Moreover, the PSF molecule uniquely combines strong compatibility with high-voltage NCM88 cathode and optimal steric hindrance, enabling a highly AGG-dominated solvation structure that its FSO_2-containing analogues with different carbon backbones fail to achieve (Figure S13, Supporting Information). Furthermore, Li||NCM811 ($LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$) pouch cells (Figure 5e inset) were assembled to evaluate the DVA electrolyte under practical cell conditions.[3,35] The cell was cycled at a charge/discharge rate of 1C/1C. The cell delivered an initial capacity of ≈548 mAh and a high capacity retention of ≈76.5% after 150 cycles, further verifying the stable CEI constructed in our DVA electrolyte even at harsh practical conditions. It clearly guides us to further investigate the interphasial characteristics in both electrolytes.

The excellent cycling stability of the NCM88 cathode at a high cut-off voltage of 4.6 V indicates that the DVA electrolyte effectively passivates the cathode surface against degradation, motivating further investigation of the associated side reactions. We performed an accelerated degradation experiment by holding the cathode at 4.6 V versus Li/Li^+ and monitoring the leakage current, which serves as an indicator of the overall side reactions (**Figure 6a**). Compared with the reference carbonate electrolyte, which stabilizes at a quasi-steady leakage current of ≈11.0 µA after an initial ≈5 h decay, the DVA electrolyte shows a continuous monotonic decrease, reaching ≈4.1 µA at the end of the 20 h hold (Figure 6a). This sustained decline reflects a diminishing side-reaction rate and effective passivation of the cathode surface. To enable a fair post-cycling comparison of the cathodes, a Li||NCM88 cell with low cathode loading (≈2 mg cm^{-2}) was cycled 100 times at 100% depth of discharge (DOD) in the carbonate reference electrolyte (Figure S14, Supporting Information), and the cathode was then extracted for analysis and compared with a cathode cycled the same number of times in the DVA electrolyte. The reduced side reactions are evidenced by two complementary observations. First, inductively coupled plasma mass spectrometry (ICP-MS, Figure 6b) reveals markedly suppressed transition-metal (TM) dissolution: after 100 cycles, the dissolved Ni content is reduced by fourfold and the total transition metals by thirteenfold compared with the reference electrolyte.

Second, in situ differential electrochemical mass spectrometry (DEMS) shows significantly lower CO_2 evolution above ≈4.3 V (Figure 6c) than in the reference cell (Figure 6d) during charging to 4.6 V versus Li^+/Li. The CEI formed in the DVA electrolyte was observed to be more uniform and thinner than that formed in the carbonate electrolyte (Figure S15, Supporting Information), indicating that the DVA can effectively passivate the cathode surface by forming a thinner and more ion-conductive CEI. The compositions of the CEIs derived from different electrolytes, which are another form of side reactions, were analyzed by XPS (Figure S16a–d, Supporting Information). Oxidative decomposition of the solvent molecules and salt anions on the NCM88 surface generated a fluorinated CEI that also incorporated minor amounts of nitrogen- and sulfur-containing species derived from the DVA electrolyte (Figure S16e, Supporting Information), which is consistent with the 3D reconstructed depth profiling images by TOF-SIMS (Figure S16f, Supporting Information). Such a hybrid CEI characteristic effectively passivates the NMC88 surface with the formation of uniform, thin, and ionically conductive CEI.

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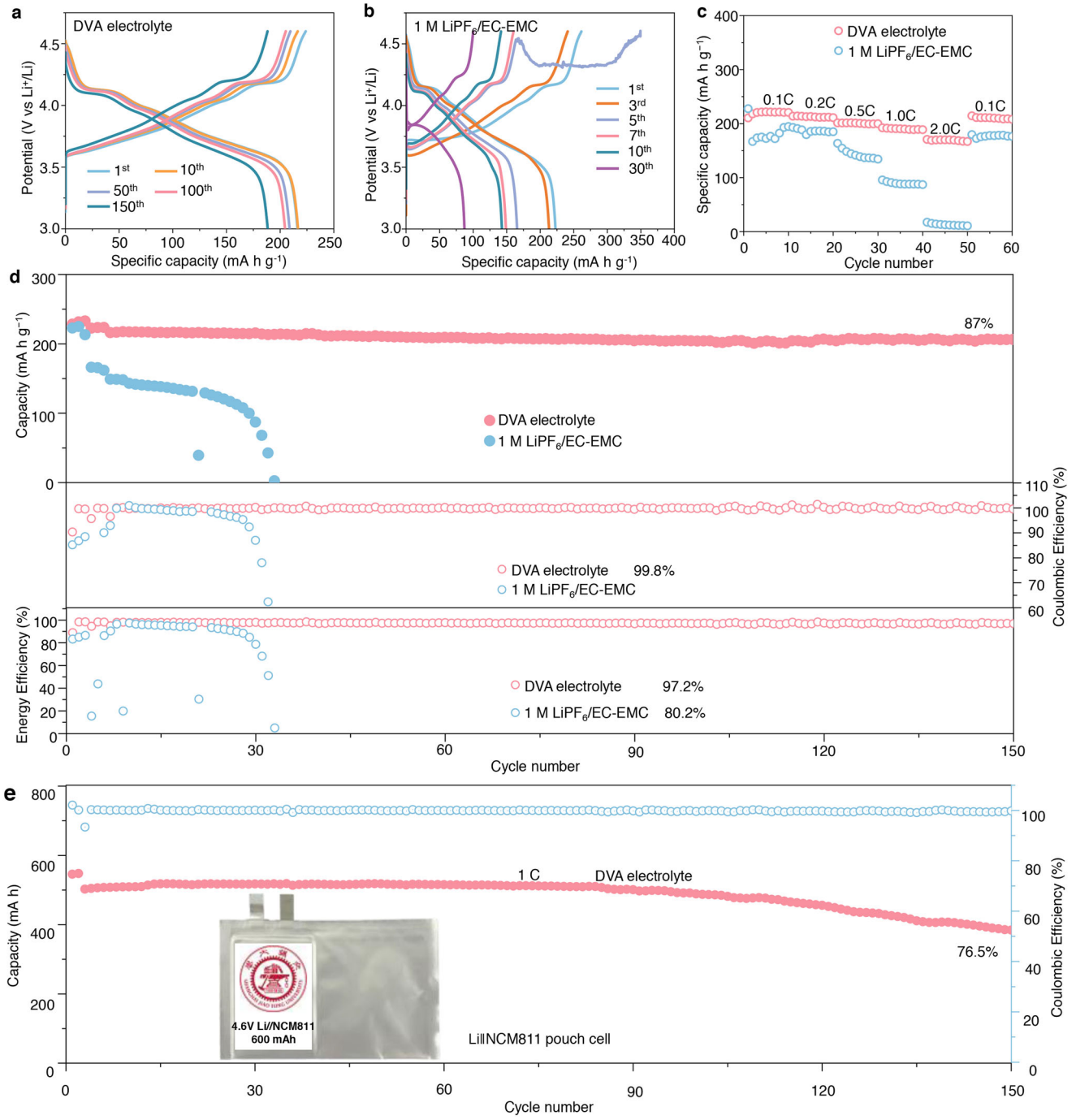


Figure 5. Electrochemical performance of high-voltage LMBs NCM88 in different electrolytes with 3–4.6 V vs. Li/Li$^+$. Galvanostatic voltage profiles of Li||NCM88 cells at 0.5 C in the DVA a) and 1 M $LiPF_6$/EC-EMC b) electrolytes. Rate c) and cycling performance with CE and energy efficiencies d) of Li||NCM88 cells with different electrolytes at 0.5 C. e) Cycling performance of the 0.6-Ah-level Li||NCM811 pouch cell with the DVA electrolyte at 1C/1C charging/ discharging.

In situ XRD analysis of the NCM88 cathode deep-charged to 4.6 V (Figure 6e) reveals the characteristic lattice evolution of Ni-rich layered oxides. During charging, the (003) reflection gradually shifts to lower 2θ values from 3.0 to ≈4.2 V, reflecting *c*-axis expansion as Li^+ extraction increases oxygen–TM repulsion and drives the H1→H2 solid-solution transition. Above ≈4.2 V the (003) peak reverses and moves sharply to higher 2θ, signifying the H2→H3 phase transition and the associated *c*-axis contraction. Other reflections, including (104), (105), and (107), show similar changes throughout the deep-charge process. Charging to 4.6 V triggers pronounced phase transitions that impose excessive anisotropic lattice expansion and contraction,

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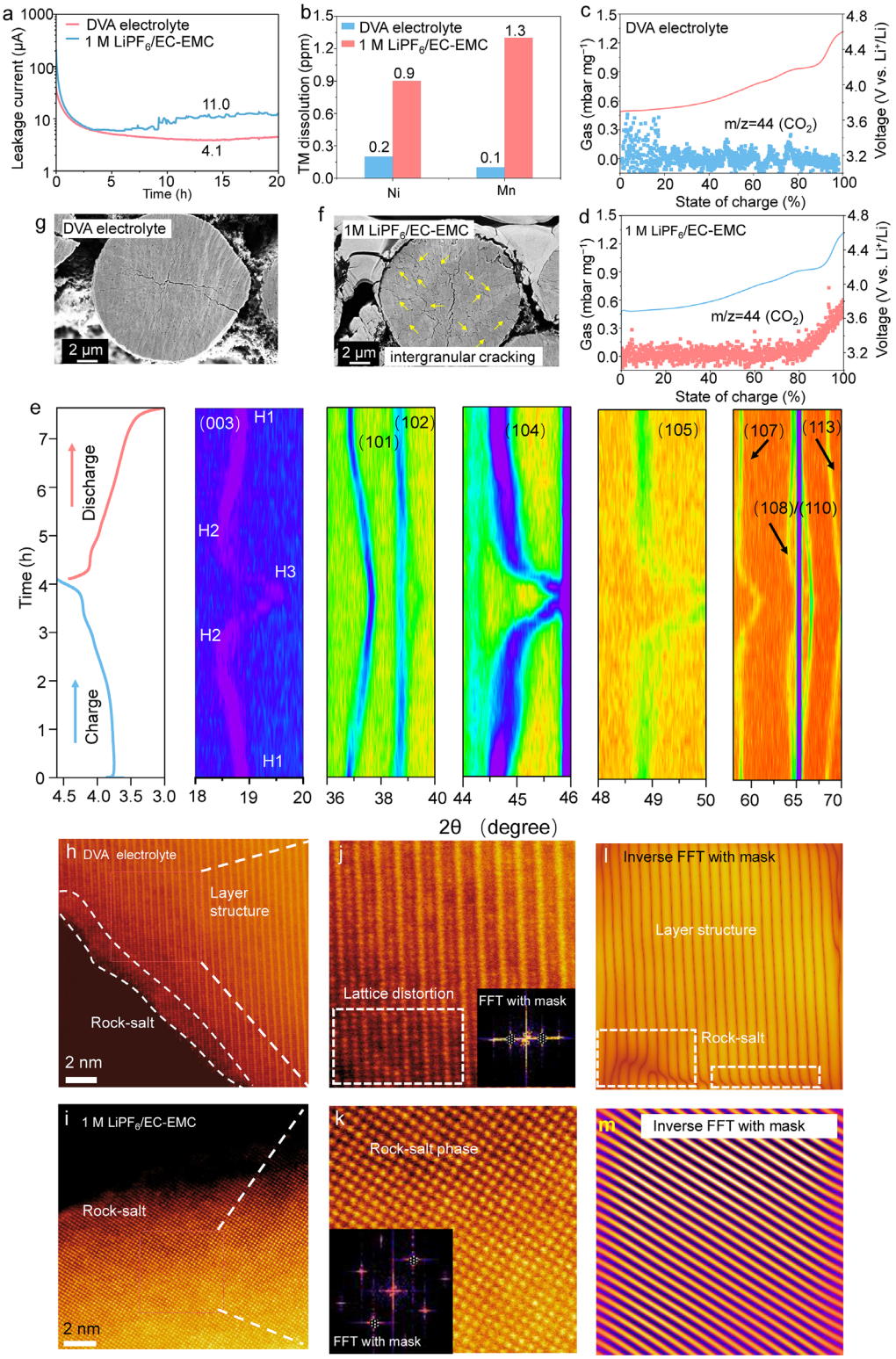


Figure 6. Degradation mechanism of the NCM88 cathode in different electrolytes. a) Leakage currents during 4.6 V constant-voltage. b) Transition metal dissolution after 100 cycles in different electrolytes. Gas evolution during first charging in DVA electrolyte c) and 1 M $LiPF_6$/EC-EMC electrolyte d). e) In situ XRD patterns from open circuit voltage to 4.6 V with corresponding voltage profiles. Cross-sectional SEM images of NMC88 cathodes after cycling in 1 M $LiPF_6$/EC-EMC f) and DVA electrolyte g) electrolyte. The yellow arrows point out the intergranular cracks within the cycled cathode secondary particle. h,i) HRTEM images, j,k) FFT, and l,m) inverse FFT patterns of the selected areas from the cathode cycled in DVA j,l) and 1 M $LiPF_6$/EC-EMC k,m) electrolyte.

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frequently initiating intergranular cracks between primary particles. Repeated cycling exacerbates this damage: the cyclic mechanical stress, combined with chemical attack from corrosive side-reaction byproducts, accelerates crack propagation through a classic stress-corrosion cracking (SCC) mechanism.[23] In this scenario, a "good" electrolyte that effectively suppresses continuous side reactions can prevent oxygen loss and TM dissolution, thus eliminating the thermodynamic and kinetic driving forces for SCC initiation and propagation and preserving the structural integrity of the cathode. Consequently, extensive cracking observed in the cathode cycled with the reference electrolyte (Figure 6f) is markedly suppressed or delayed in our electrolyte (Figure 6g). In addition to the bulk degradation of the secondary-particle agglomerates, the formation of a resistive, high-impedance rock-salt layer on the single-crystal particle surface constitutes another critical degradation mechanism, which can be further characterized by spherical aberration corrected transmission electron microscope (AC-TEM). While the rock-salt layer on the NCM88 surface cycled in the DVA electrolyte remains thin (≈2 nm, Figure 6h) and uniformly distributed, the layer on the particle cycled with the reference electrolyte grows very thick (>10 nm, Figure 6i). The underlying lattice structure is more clearly discernible in the enlarged local-area images (Figure 6j,k) and their corresponding inverse fast Fourier transform (FFT) reconstructions (Figure 6l,m). The thick, inert rock-salt layer, formed by oxygen loss and TM dissolution[36] during side reactions between the highly oxidative cathode surface and the electrolyte, blocks Li^+ diffusion paths, creating local concentration gradients that generate uneven mechanical stress due to the lattice mismatch between rock-salt and layered structure. The large mechanical stress generated during deep charge–discharge cycling, combined with corrosive species produced by the decomposition of EC on the high-valence Ni surface and its reaction with $LiPF_6$,[37] promotes intergranular SCC. In contrast, our DVA electrolyte effectively suppresses the detrimental side reactions and SCC, analogous to our previous work with the DMTMSA molecule,[23,38,39] which bears two methyl groups attached to the nitrogen atom. This outcome indicates that our PSF molecular design successfully inherits the exceptional oxidation resistance of its predecessor, DMTMSA, but also boosts the ionic conductivity from ≈1 mS cm^{-1} (1 m LiFSI in DMTMSA) to 4.9 mS cm^{-1} (1 m LiFSI in PSF).

3. Conclusion

In summary, we report a novel vehicular AGG mechanism in dilute electrolyte (1 m) based on a designed PSF solvent, integrated charge-delocalized FSO_2^- polar end, and an N-pyrrolidine ring with optimal steric hinderance, that forms AGG-dominated solvation structures to address the key challenges in high-energy LMBs. Unlike conventional solvents containing highly polar C═O or overly delocalized $-SO_2CF_3$ groups, the $-SO_2F$ moiety in PSF molecules provides balanced polarity and moderate solvation capability, facilitating abundant AGGs capable of efficient vehicular transport with high ionic conductivity of 4.9 mS cm^{-1}. This unique electrolyte facilitates the formation of a robust, rich-inorganic SEI layer dominated by the decomposition of AGG to exhibit an excellent average Li-metal CE of ≈99.5%. When paired with high-voltage NMC88 cathodes (4.6 V), the LMB can achieve 87% capacity retention after 150 cycles and good rate performance of 2C. Furthermore, our electrolyte effectively suppresses the stress-corrosion cracking, transition-metal dissolution, CO_2 gas emission, and detrimental surface degradation on the cathode side. Under practical conditions, a 0.6-Ah-level pouch cell demonstrate exceptional cycling stability, maintaining 76.5% capacity retention over 150 cycles. This study advances the fundamental understanding of solvation structures in dilute electrolytes and proposes a strategic approach for developing electrolytes for high-energy LMBs.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

W.X. led the project and conceived the concept. X.H. performed the experiments, collected the data, and completed the experimental results analysis, with the assistance of L.X., N.L., and Z.Z. X.H. conducted the DFT calculation and analysis. C.W. and Q.W. conducted the MD simulations and analysis. W.X., X.H., Y.L., C.W., and Q.W. wrote, revised, and finalized the manuscript. X.H., Y.L., and W.X. administered the funding for the projects. All the authors contributed to the entire results discussion.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

dilute electrolyte, 4.6-V high-voltage cathode, Li-metal anode, AGG-dominated solvation structure

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